

DSA 3105: Network Science Theory

MOD 11: Dynamics on Networks II: Epidemic Spreading

Learning Objectives

- Motivation and Goals
- Network Medicine: Complex Systems in Human Disease and Therapeutics
- Statistical Network Models
- Role of Network Science
- Mechanistic Models
- Exponential Random Graph Models: ERGMs
- Stepped Wedge Randomized Design
- Molecular Analysis
- Identifying Likely Transmission Pairs
- Cluster Randomized Trial
- Adaptive Rank vs Fixed Rank

The Role of Network Science in Design, Monitoring and Analysis of Cluster Randomized Trials for Control of Spread of Viral Infections

Motivation

- “The best way to think about B.1.1.7 and other variants is to treat them as separate epidemics. We’re ...obscuring the view by adding them all up to give an overall number of cases.”*
- UK guidance recommends basing approval of booster vaccines with “strain changes” on results of neutralizing antibody tests –and safety without requiring evidence of **clinical efficacy**.
- **To obtain information of the effect of different boosters on forward transmission of different major variants, we should consider new approaches that combine across clinical trials methods, molecular epi, and network science.**
- **Goal:** *evaluate how results of neutralization assays associated with different vaccines predict prevention of spread of variants.*
- May need mix of randomized and observational evidence.
- New developments in network methods may aid in design, monitoring and analysis of study results.

*Sebastian Funk, professor of infectious disease dynamics at the LSHTM.

Network Medicine: Complex Systems in Human Disease and Therapeutics

- Big data, genomics, and quantitative approaches to network-based analysis are combining to advance the frontiers of medicine as never before.
- Network Medicine introduces this rapidly evolving field of medical research, which promises to revolutionize the diagnosis and treatment of human diseases.
- With contributions from leading experts that highlight the necessity of a team-based approach in network medicine, this definitive volume provides readers with a state-of-the-art synthesis of the progress being made and the challenges that remain.
- Medical researchers have long sought to identify single molecular defects that cause diseases, with the goal of developing silver-bullet therapies to treat them.
- But this paradigm overlooks the inherent complexity of human diseases and has often led to treatments that are inadequate or fraught with adverse side effects.

- Rather than trying to force disease pathogenesis into a reductionist model, network medicine embraces the complexity of multiple influences on disease and relies on many different types of networks: *from the cellular-molecular level of protein-protein interactions to correlational studies of gene expression in biological samples.*
- The authors offer a systematic approach to understanding complex diseases while explaining network medicine's unique features, including the application of modern genomics technologies, biostatistics and bioinformatics, and dynamic systems analysis of complex molecular networks in an integrative context.
- By developing techniques and technologies that comprehensively assess genetic variation, cellular metabolism, and protein function, network medicine is opening up new vistas for uncovering causes and identifying cures of disease.

Bursts

- Bursts: is about...
- *The Hidden Pattern Behind Everything We Do...*
- *A revolutionary new theory showing how we can predict human behavior.*
- Can we scientifically predict our future?
- Scientists and pseudo scientists have been pursuing this mystery for hundreds and perhaps thousands of years.
- But now, astonishing new research is revealing patterns in human behavior previously thought to be purely random.
- Precise, orderly, predictable patterns...
- Albert-László Barabási, already the world's preeminent researcher on the science of networks, describes his work on this profound mystery in *Bursts*, a stunningly original investigation into human nature in the light of Big Data.

- His approach relies on the digital reality of our world, from mobile phones to the Internet and email, because it has turned society into a huge research laboratory.
- All those electronic trails of time stamped texts, voicemails, and internet searches add up to a previously unavailable massive data set of statistics that track our movements, our decisions, our lives.
- Analysis of these trails is offering deep insights into the rhythm of how we do everything.
- **His finding?**
- We work and fight and play in short flourishes of activity followed by next to nothing.
- The pattern isn't random, it's "bursty."
- Randomness does not rule our lives in the way scientists have assumed up until now.

- Illustrating this revolutionary science, Barabási artfully weaves together the story of a 16th century burst of human activity—a bloody medieval crusade launched in his homeland, Transylvania—with the modern tale of a contemporary artist hunted by the FBI through our post 9/11 surveillance society.
- Barabási's astonishingly wide range of examples from seemingly unrelated areas include how dollar bills move around the U.S., the pattern everyone follows in writing email, the spread of epidemics, and even the flight patterns of albatross.
- Bursts reveals what this amazing new research is showing us about where individual spontaneity ends and predictability in human behavior begins.
- The way you think about your own potential to do something truly extraordinary will never be the same.

Statistical Network Models

- Advances in statistical models for network generation allow for ***mechanistic models to be expressed as statistical models.***
- This facilitates making use of information—**including estimates of network properties and variability in them**--to simulate epidemics and impact of interventions on them.
- It also allows for inference on networks—and processes on them (e.g. transmission with or without vaccines)--with incomplete or uncertain information.

Role of Network Science

- Construction of **agent-based models** used to guide design and monitoring of studies.
- **Example:** evaluation of the impact of interference across **randomized units**—which impacts power and interpretation of results-- based on models of propagation of virus on transmission networks.
- The design of the **Botswana Combination HIV Prevention Study** was based on a simulation of interference across 30 village randomized to intervention vs control.*
- Could also aid in ascertaining error in attribution of transmissions, by making use of genetic sequence, geographic proximity and residence data to ascertain plausibility of transmission.

*Wang et al, Clinical Trials, 2014

Statistical Issues

- i. Dealing with uncertainty in contact networks and transmission networks
- ii. Incomplete data
- iii. Combining evidence

Mechanistic Models

- Models where networks are **grown or evolved over time by repeatedly applying a collection of stochastic rules** (e.g. governing edge formation) at the individual node level.
 - Rules are typically domain specific and codify the essential organizing principles of the system from the point of view of individual nodes
 - Simple rules can give rise to rich and complex structure at the **macroscopic level**

Exponential Random Graph Models: ERGMs

- Exponential-family random graph models (ERGMs) are a general class of models based on exponential-family theory for specifying the tie probability distribution for a set of random graphs or networks.
- Within this framework, one can—among other tasks:
 - i. Define a model for a network that includes covariates representing features like nodal attribute homophily, mutuality, triad effects, and a wide range of other structural features of interest;
 - ii. Obtain maximum-likelihood estimates for the parameters of the specified model for a given data set;
 - iii. Test the statistical significance of individual coefficients, assess models for convergence and goodness-of-fit and perform various types of model comparison; and
 - iv. Simulate new networks from the underlying probability distribution implied by the fitted model.

The general form for an ERGM

- ERGMs are a class of models, like linear regression or GLMs.
- The general form of the model specifies the probability of the entire network (on the left-hand side), as a function of terms that represent network features we hypothesize may occur more or less likely than expected by chance (on the right-hand side).
- The general form of the model can be written as:
- $$P(Y = y) = \exp(\theta' g(y)) / k(\theta)$$

where

- Y is the random variable for the state of the network (with realization y),
- $g(y)$ is a vector of model statistics (“ERGM terms”) for network y ,
- θ is the vector of coefficients for those statistics, and
- $k(\theta)$ represents the quantity in the numerator summed over all possible networks (typically constrained to be all networks with the same node set as y).

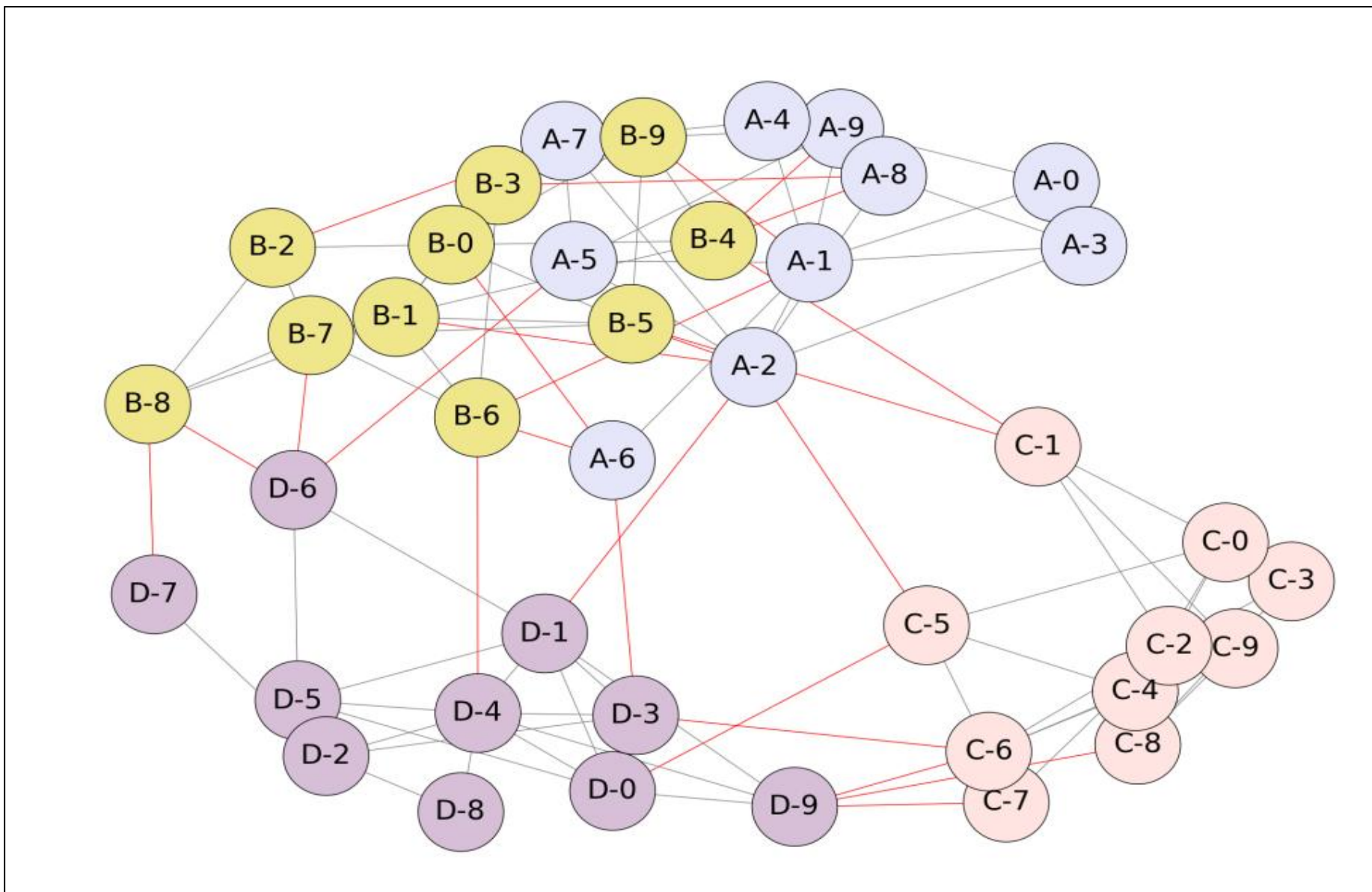
The model statistics $g(y)$: ERGM terms

- The statistics $g(y)$ can be thought of as the “covariates” in the model.
- In the network modeling context, these represent network features like density, homophily, triads, etc.
- In one sense, they are like covariates you might use in other statistical models.
- But they are different in one important respect:
 - i. these $g(y)$ statistics are functions of the network itself
 - ii. – each is defined by the frequency of a specific configuration of dyads observed in the network
 - iii. – so they are not measured by a question you include in a survey (e.g., the income of a node), but instead need to be computed on the specific network you have, after you have collected the data.

Leveraging contact network structure in the design of cluster randomized trials

Goal is to improve public health impact of a trial of Ebola vaccine or prevention (or both) while also providing randomized evidence in support of vaccine effectiveness.

Harling et al., Clinical Trials 2017



Within-cluster connections (ties) in grey, and between-cluster, in red.

cluster A has 14 ties ; B 16; C 8; D 12; ordering of cross-cluster ties is (B, A, D, C)

Stepped Wedge Randomized Design

- We order pairs of clusters in two ways:
 - i. fixed based on baseline connectivity of all clusters; and
 - ii. adaptive, including only unvaccinated clusters in measurement of connectivity
- At 1st time point, **randomize two most-connected clusters to treatment and control status.**
- At next time point, **the control cluster from step one and the next most-connected cluster are paired, randomized**—and so on.
- For modest cost in power (e.g. from 83% to 75%), simulation studies imply that we could reduce incidence by 25%.

Models can help inform policy as well as CRT design

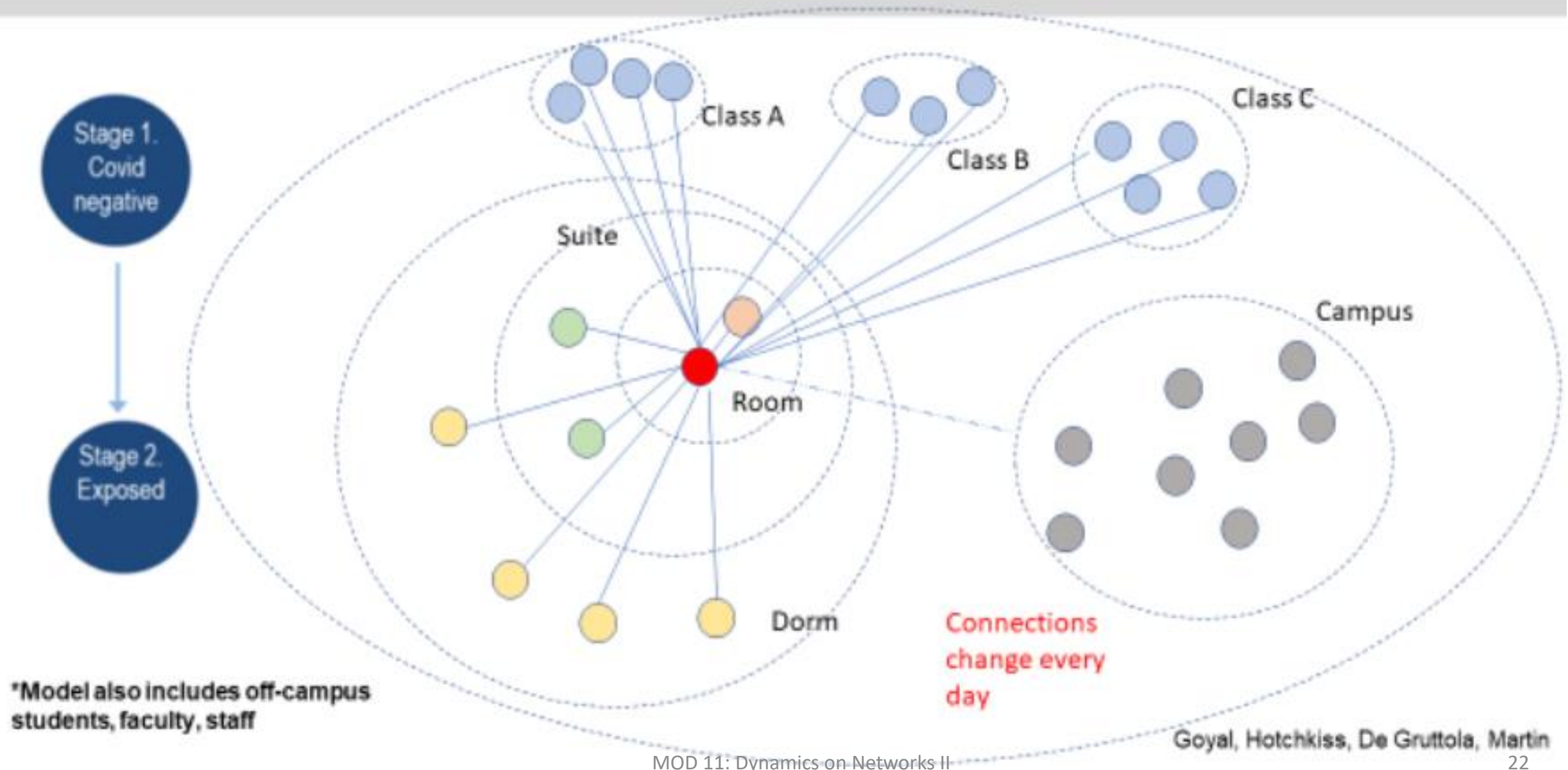
UCSD COVID NETWORK MODEL

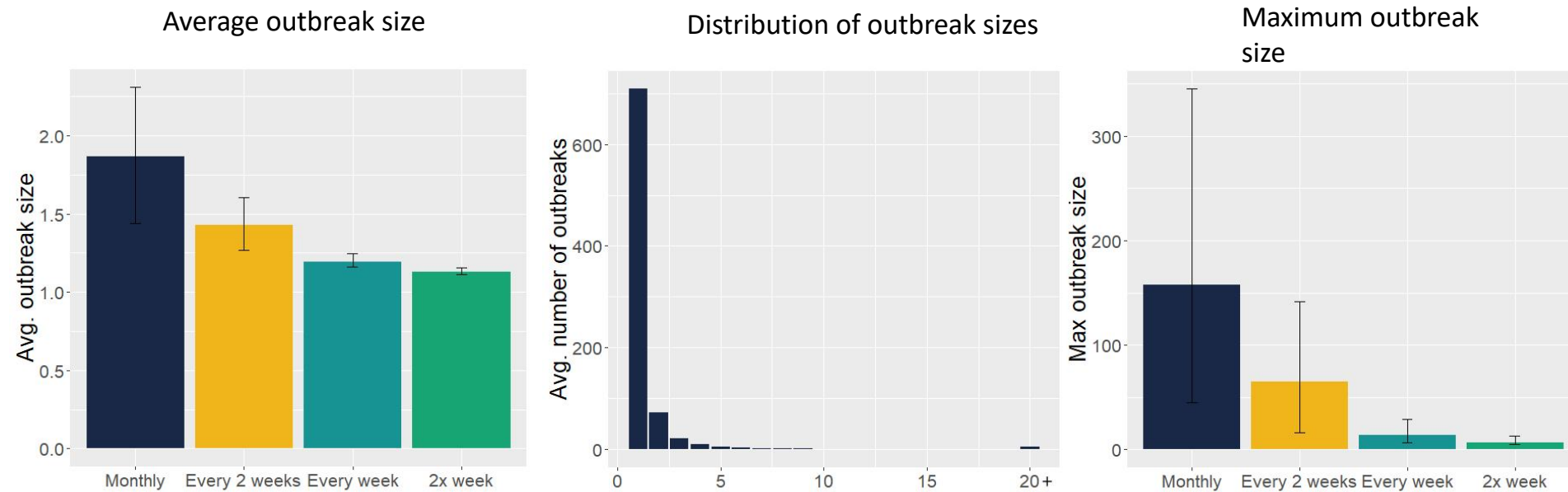
- SARS-CoV-2 transmission and natural history model
- Simulates individuals in the UCSD community:
 - Undergraduates (on/off campus, scholar athletes)
 - Graduates (on/off campus)
 - Faculty
 - Staff
- Incorporates campus-specific networks: classrooms, residential housing, general campus interactions
- Includes bolus of infections on student return and ongoing importation of infections from the outside community (from forecasting model)

Full model
schematic here



UCSD COVID NETWORK MODEL – ON CAMPUS STUDENT





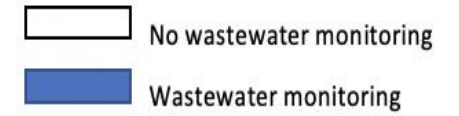
- Average outbreak size low (<2) and majority of outbreaks are small (<20).
- BUT maximum outbreak size can be large with monthly testing. Every 2-week testing reduces this to mean <80 . Weekly testing dramatically reduces the maximum outbreak size (<20)

Pragmatic CRT: UCSD Wastewater Study

WW testing can lead Dx by 5-7 days*

- **AIM:** To assess the impact of wastewater monitoring on detected cases in associated residential units and student testing behavior.
- **Unit of randomization:** clusters of manholes associated with residential buildings
- **Design:**
 - ½ clusters receive samplers first timestep
 - Remainder receive samplers second timestep
- Vertical comparisons in first time step
- Horizontal comparisons in both immediate and delayed tester arms

Phase	Pre-intervention	Roll-out	
Study Period	Weeks 1-2	Weeks 3-6	Weeks 7-10
Site 1			
Site 2			
Site 3			
Site 4			
Site 5			
Site 6			
Site 7			
Site 8			
Site 9			
Site 10			
Site 11			
Site 12			
Site 13			
Site 14			



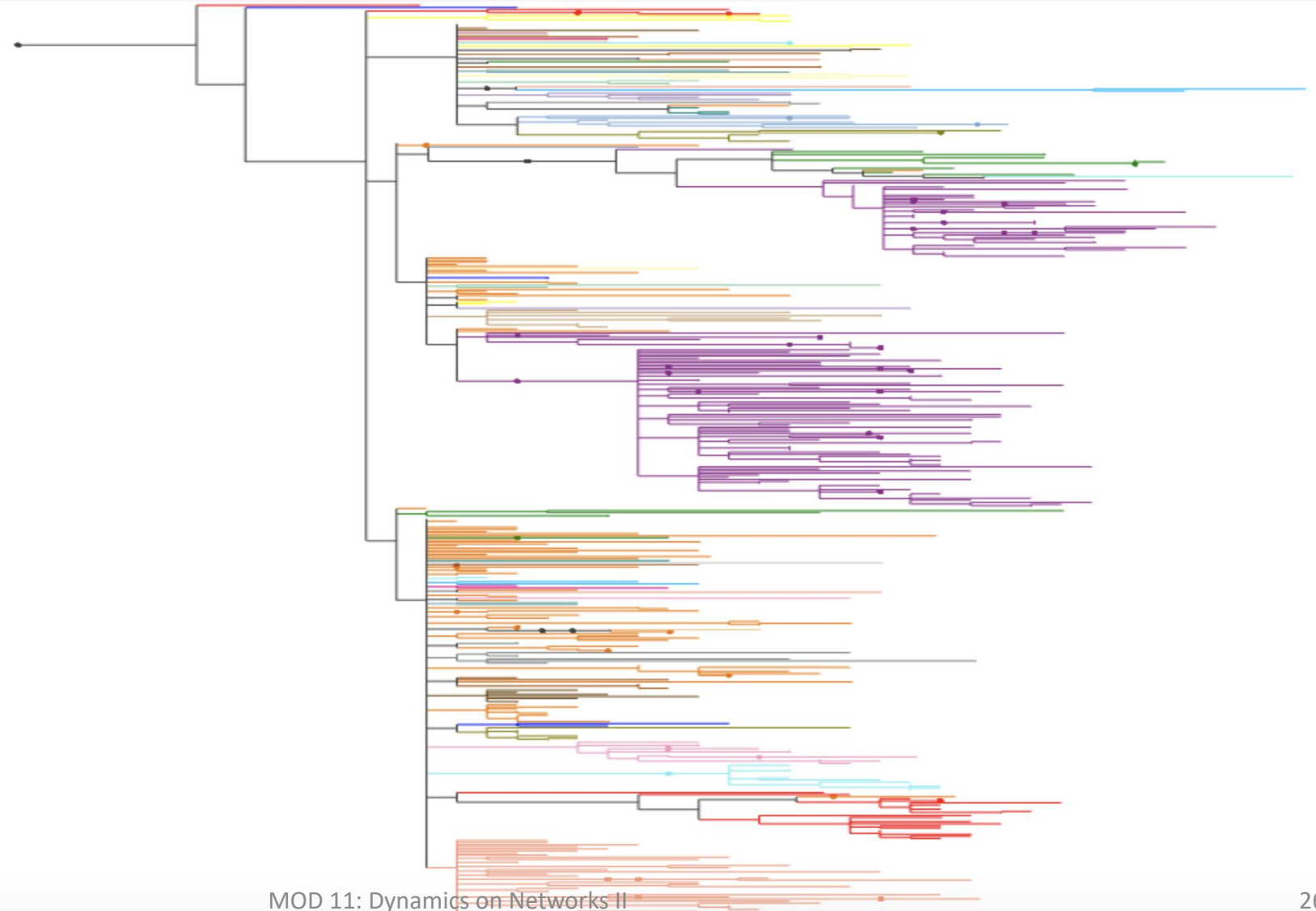
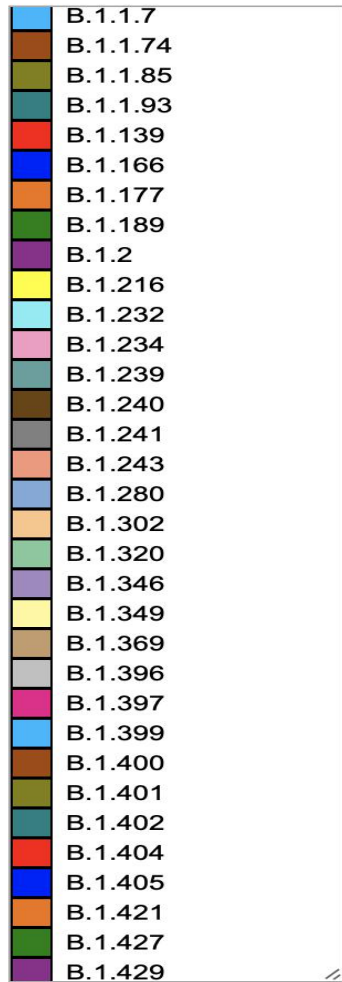
* Karthikeyan, Knight et al.

Viral genetic data

- SARS-CoV-2 mutation rate is slow—if epidemic grew from a single introduction, it would show all infections in one cluster
- But if pairs of sequences have >1 AA change between them, one can rule out that people with different variants could have infected each other.

San Diego & UCSD Coronavirus Genetics

Andersen, Knight, Laurent,
Vazquez-Baeza, Pride, Pollock,
Longhurst, Kellen, Peng, White,
Sathe, and others....



Booster Vaccine Study Design

- **Campus Study design:** randomize residential units (room, suite, floor) to two (or more) boosters: A, B,
- **Use phylogenetic analyses of campus infections to identify:**
 - **Major Variants (MV)**—those with mutations that may impact vaccine efficacy.
 - **minor variants (mv)** —nested within MV-- whose sequences are either identical or have 1 AA change.
- **Cluster randomize participating students;** collect information on whether non-participants on campus take A, B or other boosters or no booster.
- Test all on campus.

Molecular Analysis

- During study, sequence all new infections and assign each to an existing mv (minor variant) (if such exists) and to an MV (Major Variant).
- Monitor growth of clusters of people with specific mv.
- Make mv's (minor variant's) new to campus, “seeds” of a new cluster.
- Follow the growth of mv's and MV's within each arm—as well among all students/employees on campus not in the study.

Study data

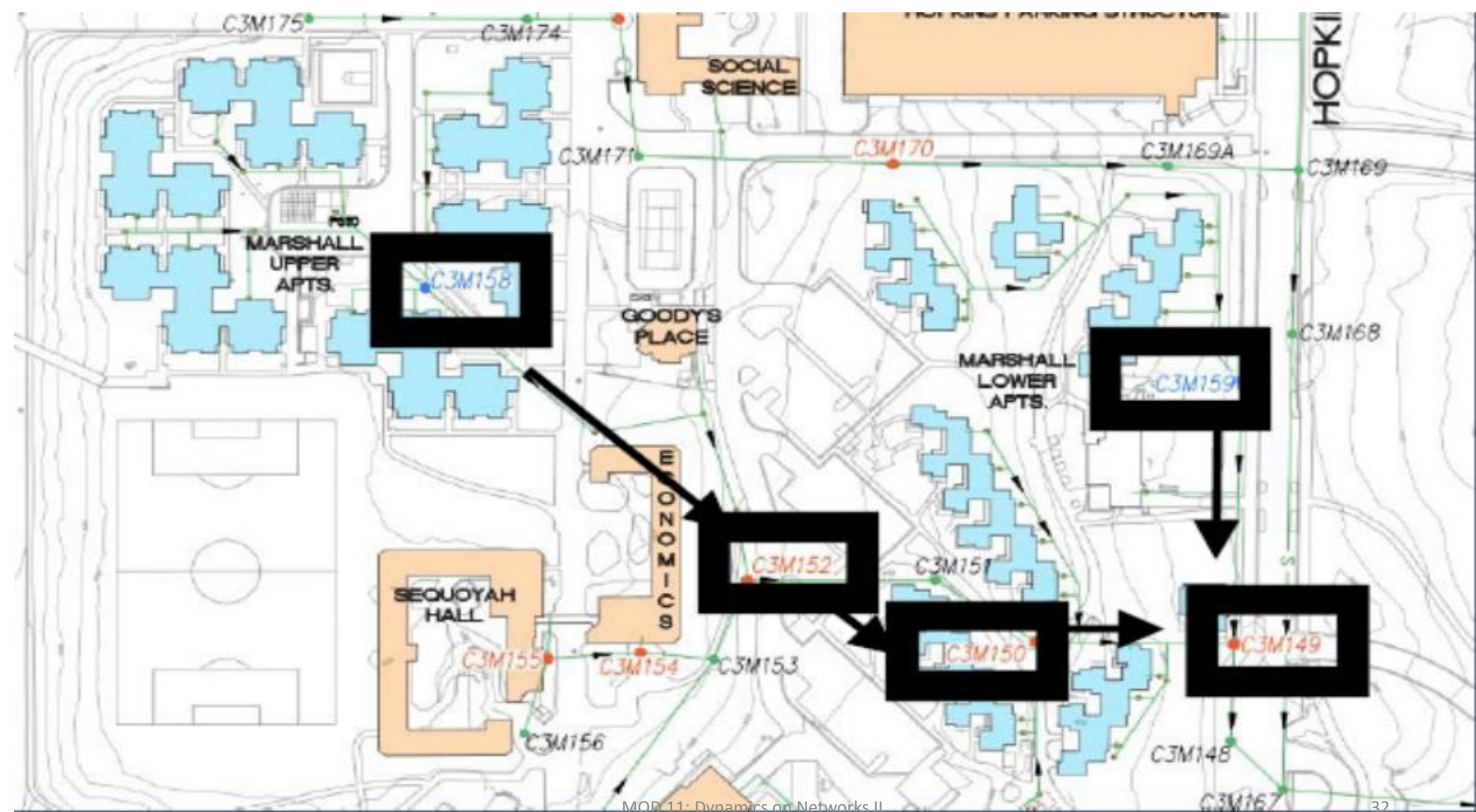
- Identify transmission pairs (with uncertainty) using the following information:
 - i. Contacts through known structures (residences, classrooms)
 - ii. Bluetooth proximity devices.
 - iii. Contact tracing done by university.
 - iv. Sequence data

Summary and Conclusions

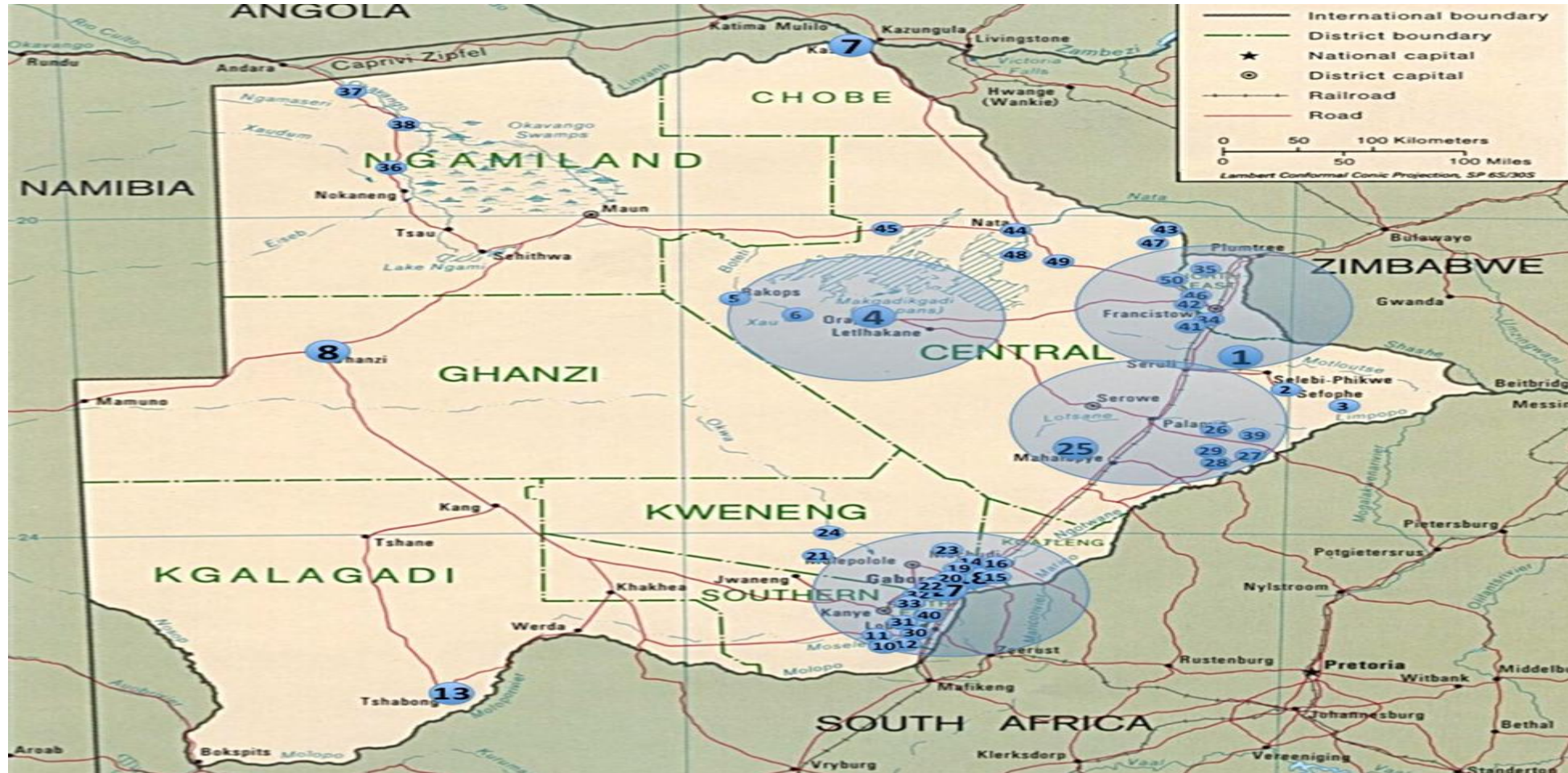
- New approaches will be needed to study impact of “strain-changed” boosters on spread of viral variants.
- Knowledge of contact networks and transmission networks—and associated uncertainty—is important for design, monitoring, and analysis of such studies.
- **New network methods make it possible to evaluate how uncertainties in network play out as uncertainties in properties of designs and results of analyses.**
- Agent-based models of microbial spread on networks aid in monitoring and in interpretation of results

Identifying Likely Transmission Pairs

- Use contact as well as **viral genetic data** to reduce set of plausible transmission pairs.
- Further reduction could be provided by an **adjudication committee**.
- Use totality of available data to **assign weights based on strength of evidence**--including not only data this study but also from other transmission studies.
- For example, ***same-sequence roommates infected a few days apart might receive higher weight than such pairs of people in same classroom.***

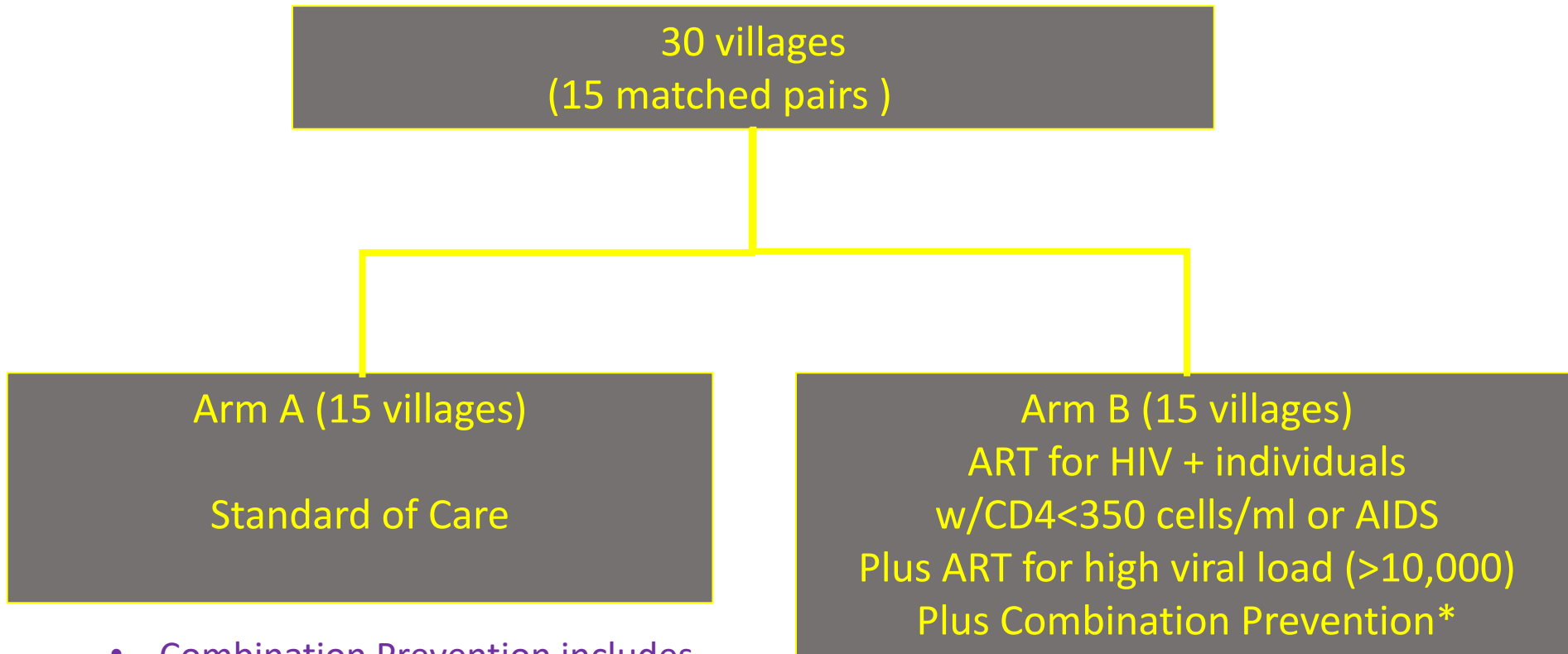


BCPP : to evaluate the impact and cost effectiveness of combination HIV prevention interventions on population-level HIV incidence in Botswana



Map of 50 Botswana communities with population sizes ranging from 3,000 to 15,000 (provided by Hermann Bussmann and colleagues)

Design Overview

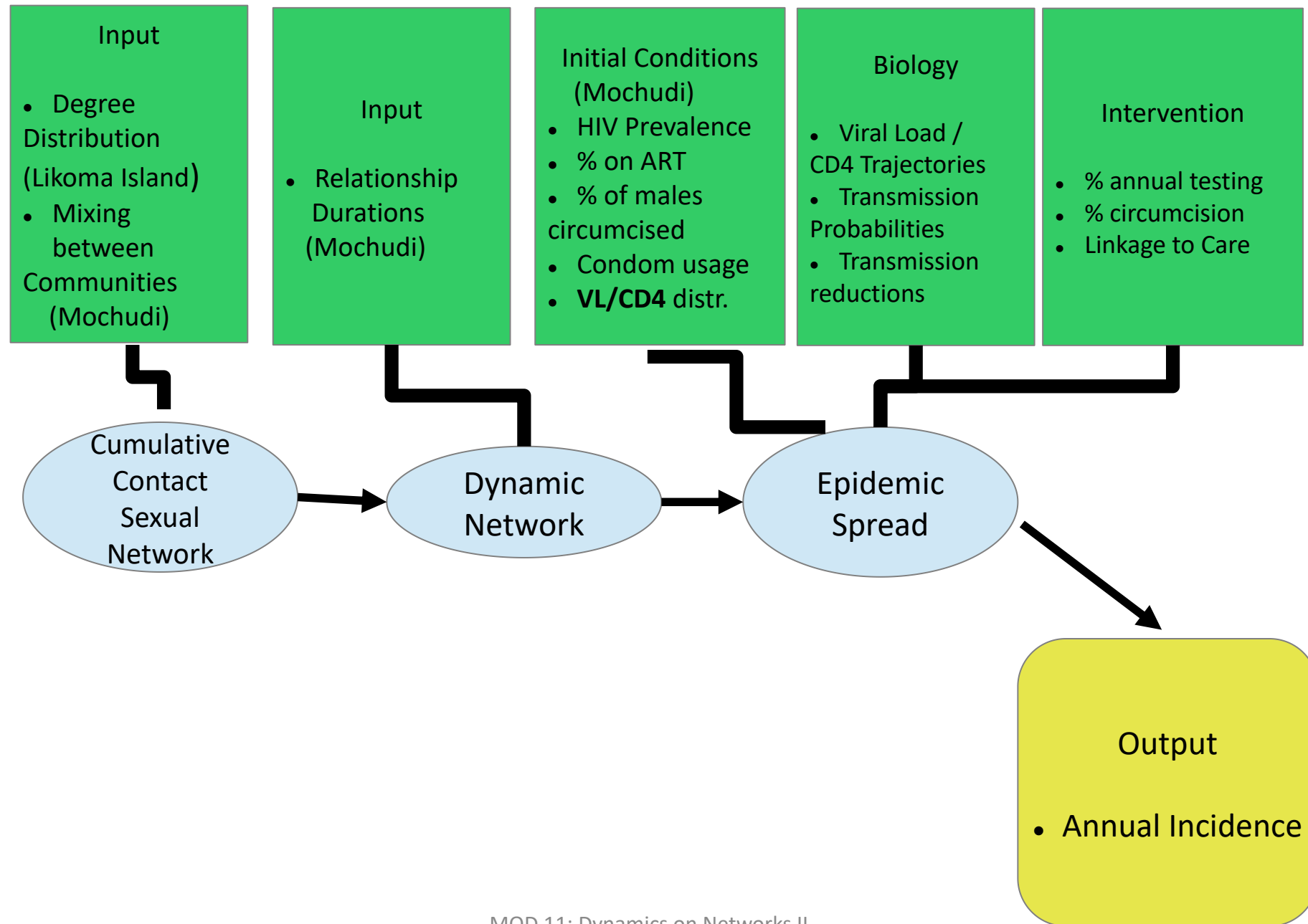


- Combination Prevention includes
 - Enhanced HIV testing and counseling (HTC),
 - Prevention of mother-to-child transmission (PMTCT),
 - Enhanced test-linked care (TLC) in relation to ART
 - Male circumcision (MC)
- Both arms will evaluate HIV incidence, from cohorts (20% of the population) followed longitudinally for 3 years.

Simulation studies

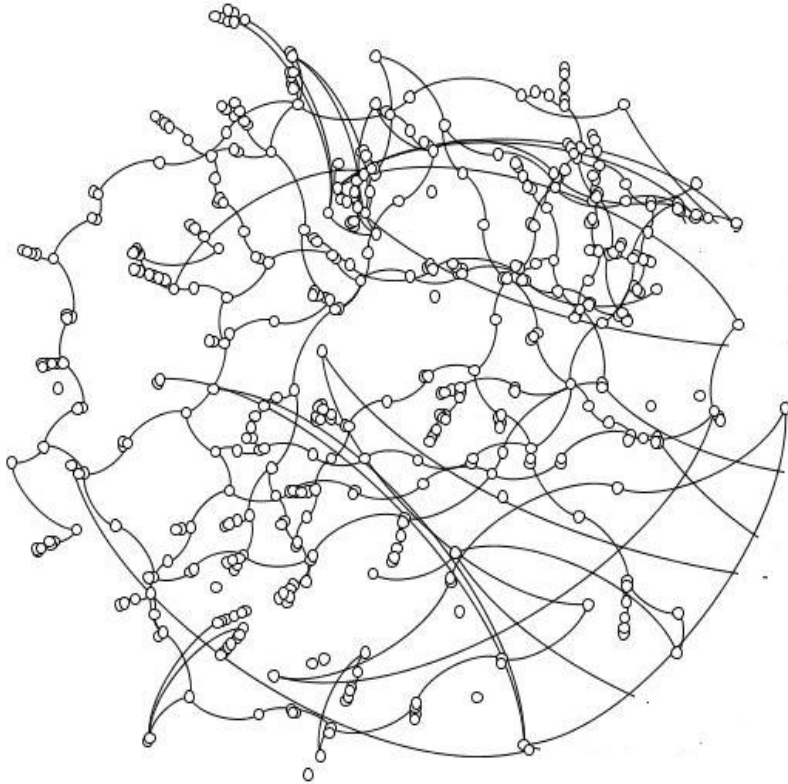
Model intervention impact on HIV spread over 4 years

- Generate **sexual networks** then propagate disease spread on these networks
- Community characteristics:
 - Sexual network characteristics (including mixing between communities)
 - Varying coverage level for different prevention modalities
 - Population sizes
- Individual characteristics:
 - i. Transmission risk
 - ii. Disease progression
 - iii. Condom use
 - iv. Linkage to care
 - v. Circumcision status

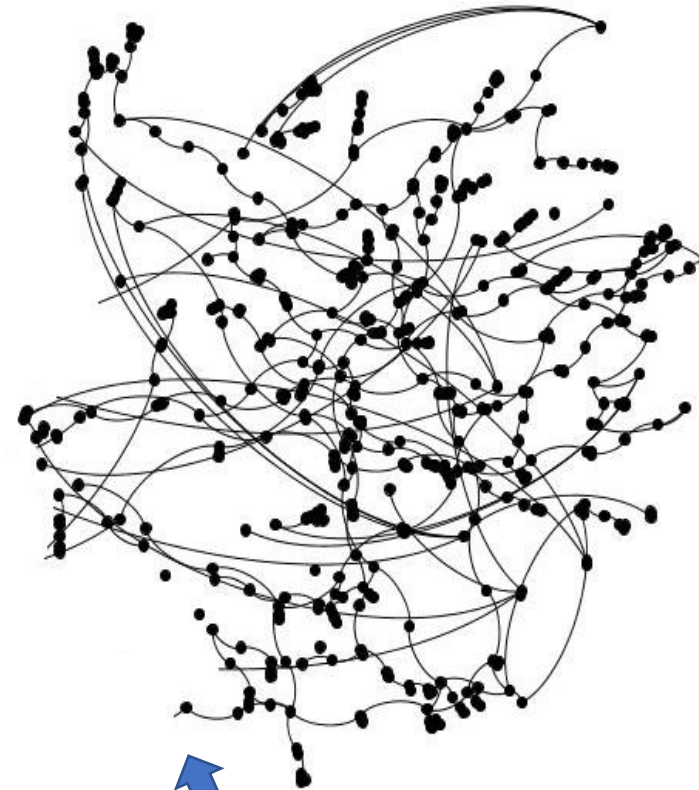


Cluster Randomized Trial

Control
Community



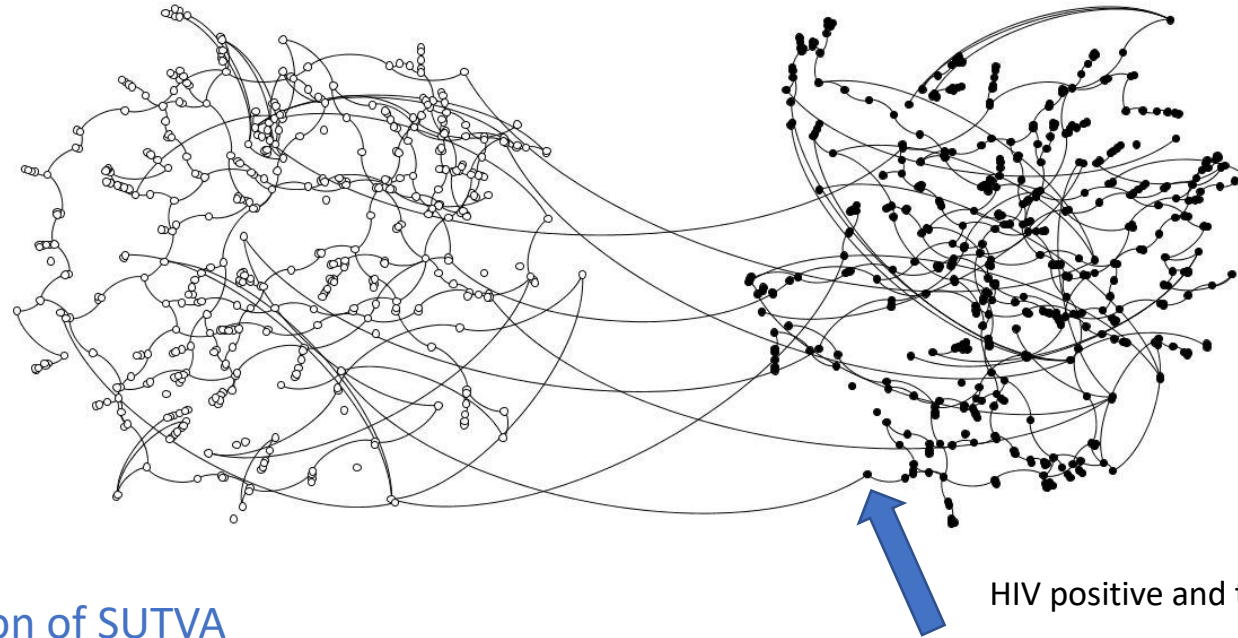
Intervention
Community



HIV positive and treated

Contamination

Control
Community

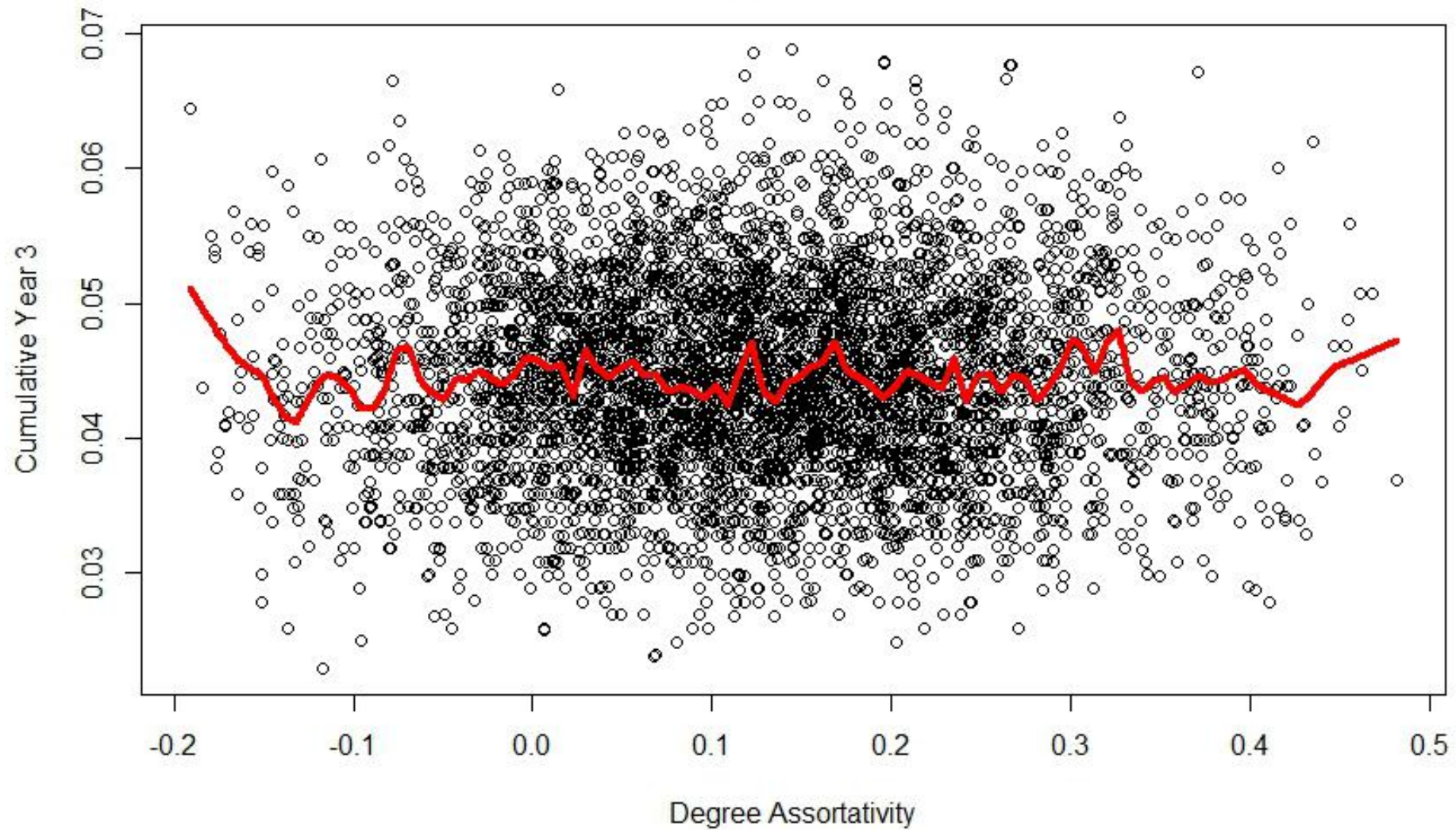


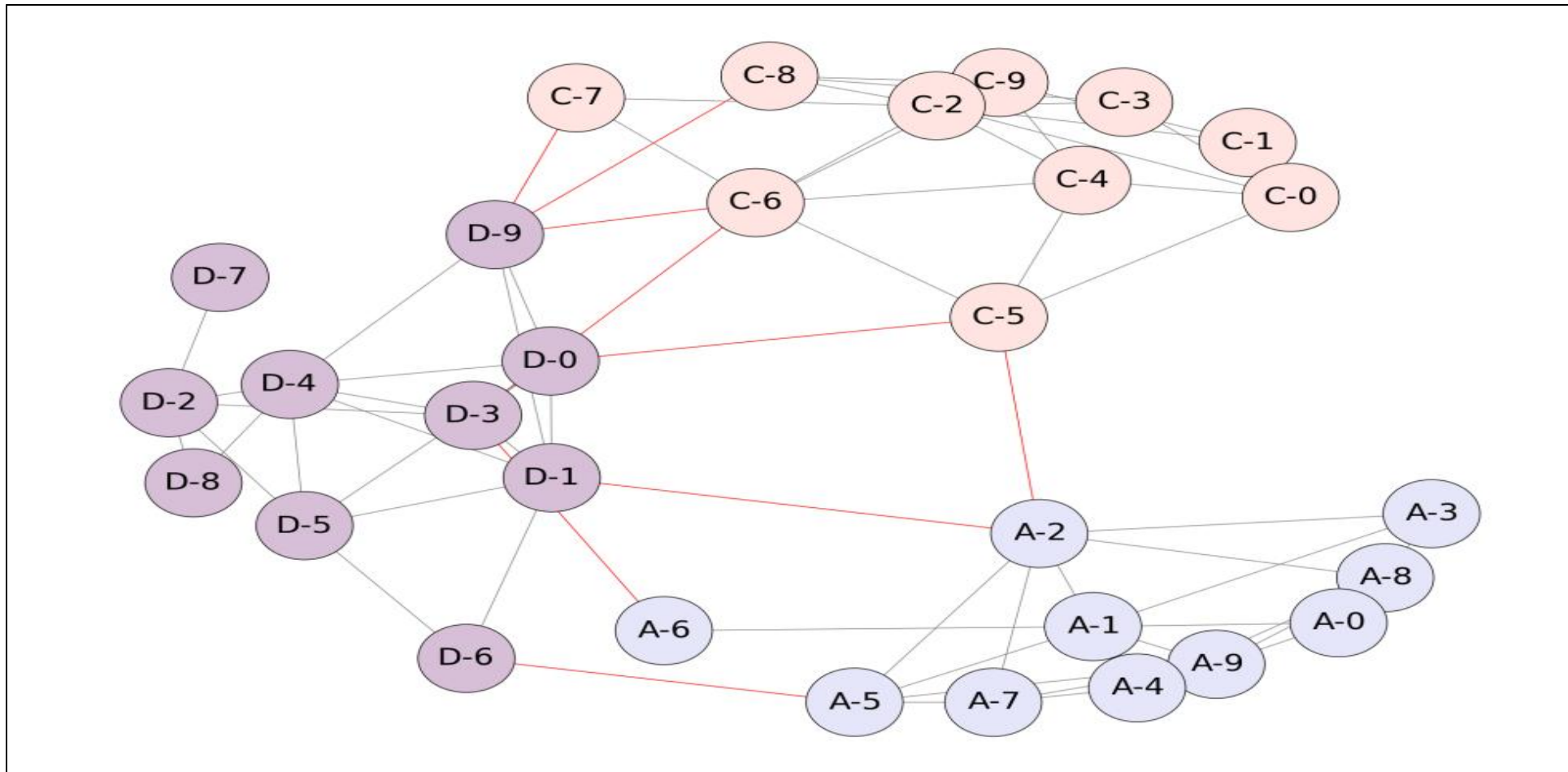
Intervention
Community

Violation of SUTVA

HIV positive and treated

Mixing by Activity and Incidence (SOC)





After cluster 'B' is treated and removed, cluster 'A' goes from second most-connected to least-connected cluster; the ordering is changed to (D, C, A).

Analysis

- Assign directionality based on difference in dates of detection (Difference must be between 2 and X days). Within 2 (?) days, directionality is not established.
- Describe each source and recipient partner in a transmission pair as: $XMVmv$, where X indicates study arm for study participant, or booster used by each non study participant.
- Denote each pair by $XMVmv_S \rightarrow XMVmv_R$.
- Denote pair for whom no directionality can be established:

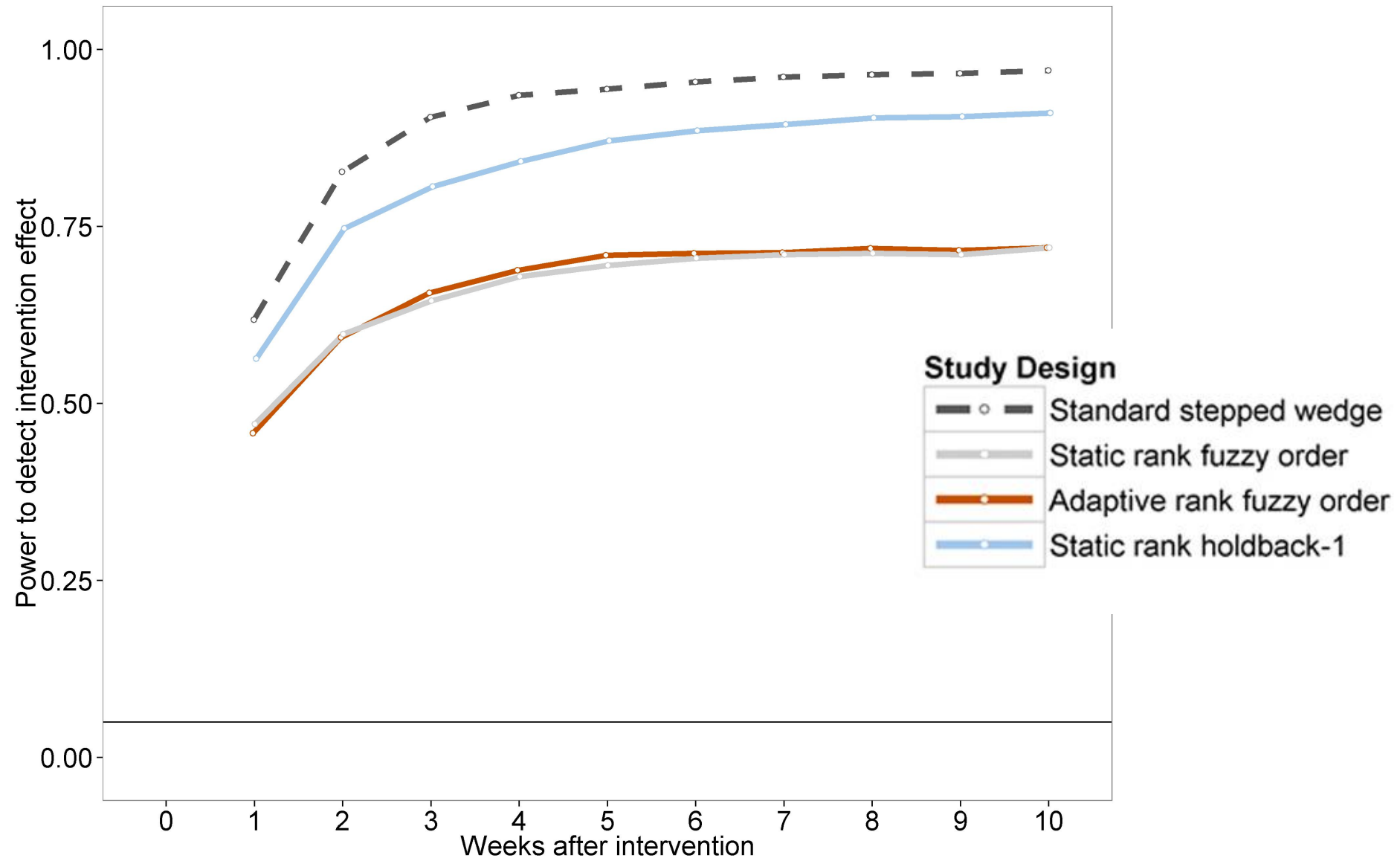
$$XMVmv_1 \leftarrow \rightarrow XMVmv_2$$

- Count the number of such pairs with different combinations of X, MV and mv. From this we can infer the total number (and rate of growth) for each MV (and mv) under different values of X

Adaptive Rank vs Fixed Rank

- Vaccinating a cluster highly connected to already-treated clusters may provide less benefit than if the cluster is highly connected to untreated clusters.
- Adaptive Rank approach includes only unvaccinated clusters in measurement of between-cluster connectivity.
- This approach preferentially targets cross-cluster ties with most potential to transmit infection.

Power



End